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To What Extent Does Discounting 'Hot' Climate Models Improve the Predictive Skill of Climate Model Ensembles?

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Abstract It depends. The Intergovernmental Panel on Climate Change's (IPCC) Assessment Report Six (AR6) took a step toward ending so-called 'model democracy' by discounting climate models that are too warm over the historical period (i.e., models that 'run hot') when making projections of global temperature change. However, the IPCC did not address whether this procedure is reliable for other quantities. Here, we explore the implications of weighting climate models according to their skill in reproducing historical global-mean surface temperature using three other climate variables of interest: global average precipitation change, regional average temperature change, and regional average precipitation change. We find that the temperature-based weighting scheme leads to an improved prediction of global average precipitation, though we show that this prediction could be overconfident. On regional scales, we find a heterogeneous pattern of error reduction in future regional precipitation. This stands in sharp contrast with the broad regional pattern of error reduction in future temperature projections, though we do find regions where error is not significantly reduced. Our results demonstrate that practitioners using weighted climate model ensembles for climate projections must take care when weighting by temperature alone, lest they produce unreliable climate projections that result from an inappropriate weighting procedure.

Plain Language Summary Climate model ensembles are widely used for risk assessment. However, a few of the most recent generation climate models 'run hot' in the historical period, widening the spread of future global warming. The Intergovernmental Panel on Climate Change's (IPCC) sixth assessment report presents a number of weighting schemes to address this 'hot model' problem, each of which discount models that are 'too hot' in the historical period. However, it is unclear if this procedure is reliable for other quantities of interest. Here we explore the impact of this procedure on global average precipitation change, regional temperature change, and regional precipitation change. We find that while this scheme improves the prediction of global precipitation change and generally improves the prediction of regional temperature, it does not broadly improve regional predictions of future precipitation change. We conclude that users of climate model output must be careful when applying a global temperature-based weighting scheme in regional impact studies.

Key Points:

- Using historical warming to weight climate models can improve global predictions of annual temperature change and precipitation change
- Using past warming to weight future climate projections has varied effects on regional error reduction depending on the metric of interest
- Climate model end-users should use caution when applying a weighting scheme to avoid biased or over-confident assessments of climate impacts

Supporting Information:

Supporting Information may be found in the online version of this article.

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1. Introduction

The Coupled Model Intercomparison Project Phase 6 (CMIP6; Eyring et al., 2016) includes numerous updates to physical processes that have substantially broadened the range of equilibrium climate sensitivity (ECS) to include values much higher than previous CMIP generations (Forster et al., 2020; Zelinka et al., 2020). These more sensitive models also simulate more end-of-century warming and have been criticized for 'running hot' (Hausfather et al., 2022). 'Hot models' in CMIP6 generally have two common biases: (a) they simulate too much warming over the last four decades because their transient climate response (TCR) is outside generally accepted values (Hausfather et al., 2022), and (b) they have an unrealistically large estimate of ECS relative to state-of-the-art estimates (Sherwood et al., 2020).

In response to this bias, the Intergovernmental Panel on Climate Change's (IPCC) Sixth Assessment Report (AR6) started down-weighting 'hot models' when providing projections of global-mean temperature changes (Eyring et al., 2021). This model weighting method ended so-called 'model democracy' present in previous CMIP generations (Brunner et al., 2019; Knutti, 2010), in which all models are given equal weight in computing the ensemble average of a given climate variable (i.e., end-of-century temperature rise). To summarize the IPCC's approach, they used a percentile-by-percentile average of three distinct weighting schemes (Liang et al., 2020;

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Ribes et al., 2021; Tokarska, 2020), each of which discount models that are 'too hot', to form a constrained future projection of relative global surface air temperature change, and then utilized an emulator to generate future projections (see Figure 4.11 and supplemental data in Eyring et al., 2021). One method present in AR6, Liang et al. (2020), show that their weighting approach reduces overall bias in future global-mean temperature projections (via cross validation), and therefore provides a more precise estimation of *global-mean* warming.

It is unclear, however, if weighting climate model projections by historical global-mean warming trends has skill for quantities other than global-mean warming, specifically those that may not be well correlated with global temperature changes (Hausfather et al., 2022). For example, while global temperature changes have been shown to correlate with *global* precipitation changes, the correlation appears to be strong only in the polar regions (Shiogama et al., 2022), suggesting that weighted ensembles might not offer a more skillful prediction for precipitation in the mid- and low-latitudes.

AR6 itself offers no guidance for weighting quantities other than global temperature. The proposed workaround —using global warming levels—is unfortunately not informative for any estimates of impacts or risk that require time-horizons, broadly classified as 'transition risks' (Bauer, Proistosescu, & Wagner, 2024). For example, adjustment costs (Lucas, 1967; Mussa, 1977) and economic inertia (Ha-Duong et al., 1997) link the cost of abating CO_2 emissions to the rate of abatement and have been shown to imply much more aggressive near-term climate policies (Bauer, McIsaac, & Hallegatte, 2024; Campiglio et al., 2022). Therefore, the rate of warming—which directly influences the optimal rate of abatement in integrated assessment models with adjustment costs—is an important consideration for policymakers. Worse still, it is unclear if a better prediction globally implies a uniformly better prediction on regional scales, or if the bulk of the precision is gained in locations relatively uninteresting for a specific impact analysis (i.e., in polar regions, as opposed to the low- and mid-latitudes).

Here we demonstrate the issues of weighting models according to their skill in reproducing global-mean surface temperature using three other climate variables of interest: annual average precipitation, regional temperature change, and regional precipitation change. We use a weighting scheme that is most similar to Liang et al. (2020), which itself expands on the weighting scheme outlined in Knutti et al. (2017). The general approach is to compute model weights using a model's ability to replicate historical warming, while also accounting for model interdependency (see the Supporting Information S1 for more details). The ability for a given weighting scheme to reduce out-of-sample prediction error is evaluated via a perfect model test (Supporting Information S1), which can be summarized as: (a) a model is randomly chosen as truth, and referred to as the 'pseudo-observation'; (b) the other models are weighted based on their ability to reproduce the historical period in the pseudo-observation; (c) the weighted ensemble projections are compared with 21st century predictions from the model chosen as pseudo-observations. This procedure is carried out with each ensemble member as the pseudo-observation once to produce the distribution of 'perfect model test errors' seen in Figure 1 and to compute the change in Root Mean Squared Error (RMSE) in Figure 2.

2. Results

2.1. Global Analysis

We first apply the temperature-based weighting scheme to projections of 21st century global average temperature change, which is analogous to AR6's analysis. We show the distribution of perfect model test errors in Figure 1a, which reflects the distribution of error for all pseudo-observation choices. We find that applying the temperature-based weights reduces RMSE between the weighted and unweighted distribution by 25.4% and reduces the relative forecast error (RFE; the ratio of the weighted ensemble variance and the unweighted ensemble variance, see the in Supporting Information S1) by 18%. This reduction in RMSE suggests that weighting models by their ability to reproduce historical warming results in a more reliable prediction of future global-mean temperature. Likewise, a reduction in RFE implies that the spread of the weighted future projection is less than the unweighted ensemble, which naturally follows from the weighting scheme discounting models that are dissimilar to the global temperature trend of the chosen pseudo-observation. Our findings are consistent with those found in Liang et al. (2020), and corroborate the idea that historical temperature trends can be used to constrain global temperature projections.

Next, we apply the historical temperature-based weighting scheme to global precipitation projections. We find that weighting future precipitation projections by historical warming trends reduces RMSE by 17.8% and

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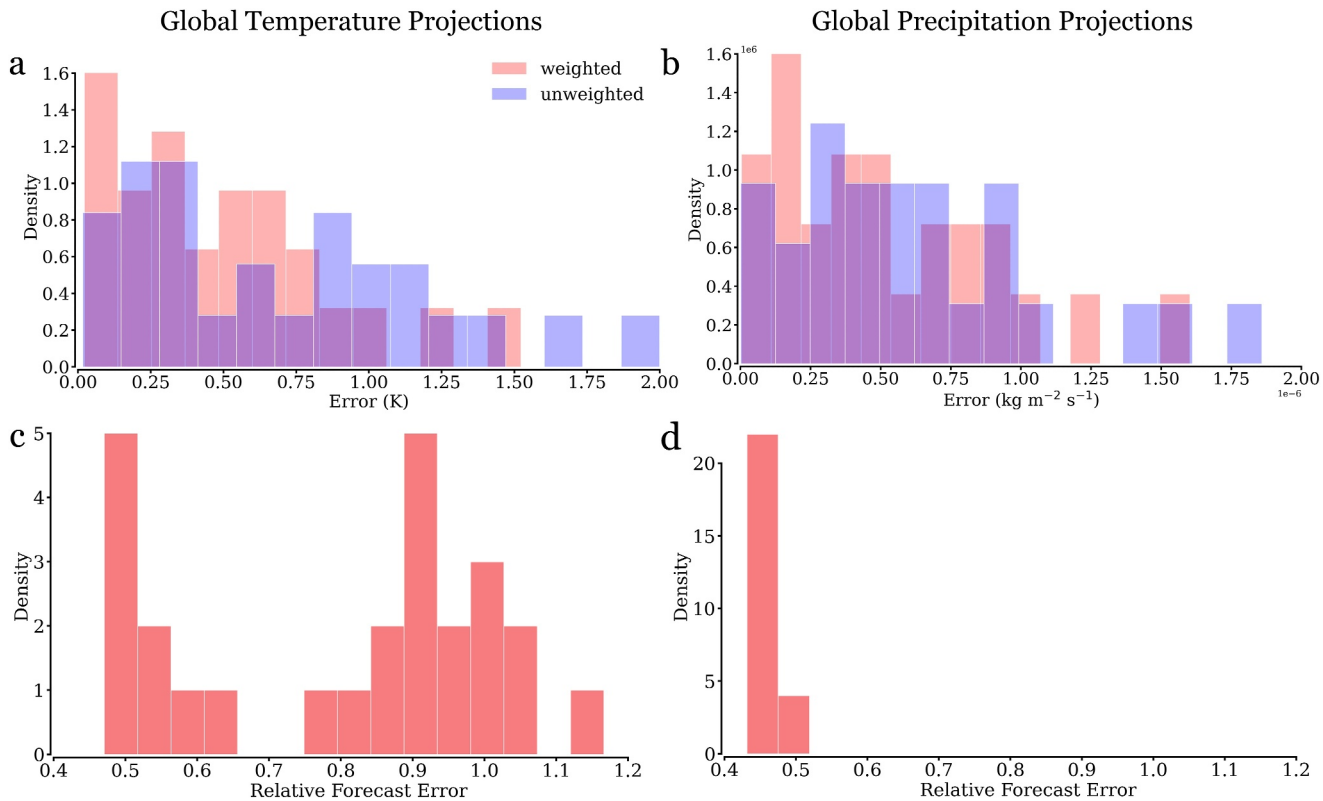

Figure 1. Perfect model test error and relative forecast error distributions. Panel a shows a histogram of the perfect model test errors in projections of 21st global warming using weighted (red) and unweighted (blue) distributions. Panel b is as Panel a, but for the error in global precipitation projections using temperature-based weights. Panel c shows the histogram of relative forecast error for global temperature projections using temperature-based weights. Panel d is as Panel c but for global precipitation projections using temperature-based weights.

decreases RFE by 55%. This suggests that temperature-based weights are useful in constraining global-mean precipitation projections, but could introduce overconfidence; indeed, the RFE is reduced three times more in the precipitation projections than in the temperature projections. The temperature-based weighting scheme likely decreases global precipitation RMSE and RFE because global temperature trends and global precipitation trends have been shown to be well-correlated (Shiogama et al., 2022). This correlation may be explained by the fact that models may agree on global precipitation changes (Held & Soden, 2006), while nonetheless disagreeing on the specific pattern of precipitation and precipitation change.

Judging by the results for the global-mean, one may be tempted to conclude that historical temperature trends are a reasonable predictor of future precipitation anomalies in CMIP6. This result is consistent with past work framing historical temperature as a possible emergent constraint on future global average precipitation projections (Shiogama et al., 2022). But does it follow that climate model practitioners will find this procedure useful in their impact analysis for, say, an individual city? To answer this question, we next look at regional decompositions of RMSE changes.

2.2. Regional Analysis

We find that weighting models by global-mean historical warming trends produces a well-defined pattern of RMSE reduction for regional warming projections (Figure 2a). This pattern can be explained by the robust pattern of correlation between the ensemble spread in *global-mean* 21st century temperature trends and the ensemble spread in *regional* 21st century temperature trends (Figure 2b). The high degree of correlation over most of the world implies that future regional temperature trends are robustly predicted by future global-mean temperature trends, which are themselves constrained by historical global-mean temperature trends.

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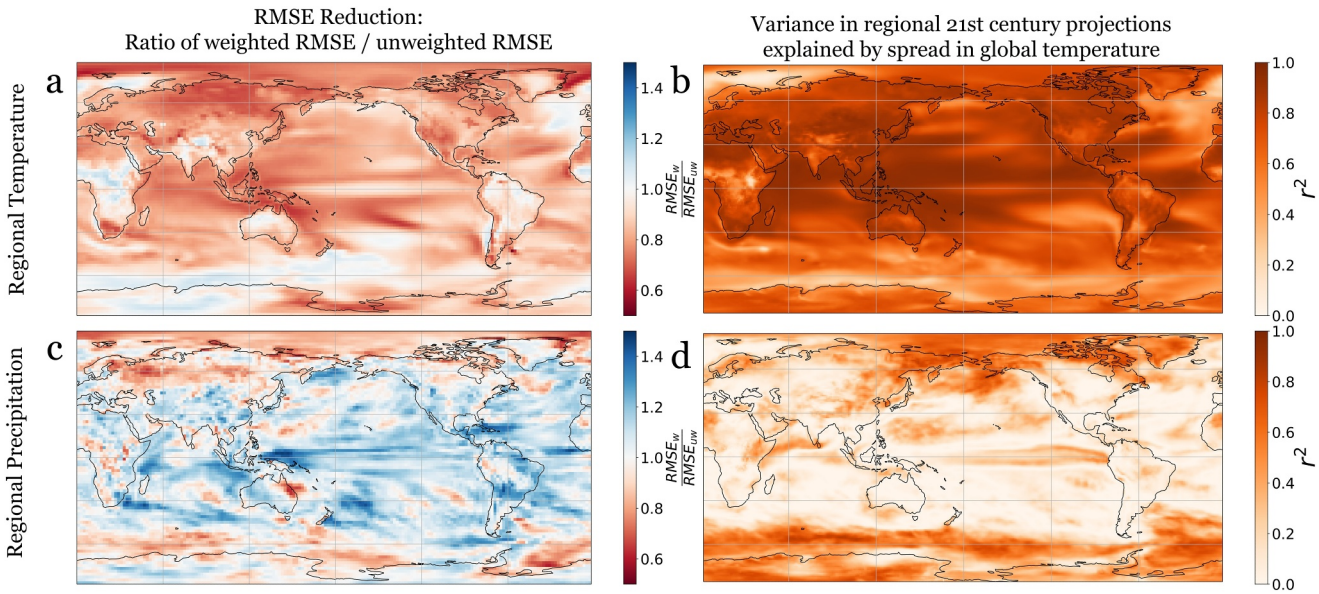

Figure 2. Regional decomposition of Root Mean Squared Error (RMSE) reduction and variance explained by weighting metric. Panel a shows the spatial distribution of the relative RMSE between the raw ensemble mean and the historical temperature trend-derived weighting technique, averaged over each pseudo-observation choice. Here, a value less than (greater than, resp.) unity implies a more (less, resp.) precise prediction using the weighted ensemble mean as opposed to the unweighted ensemble mean. Panel b shows the variance in future regional temperature anomalies that is explained by historical global temperature trends. Panel c is as Panel a, but for precipitation projections weighted by historical temperature trends. Panel d shows the variance in future regional precipitation anomalies explained by historical global temperature trends. Note that high levels of variance explained should correspond to a relative RMSE of less than one.

The locations where regional projections are least improved by weighting correspond to locations where the correlation between global and regional trends is low. These are locations where surface temperature is strongly controlled by local ocean processes, such as parts of the Southern Ocean and the North Atlantic. The implication is that in these regions the uncertainty in local warming trends is not primarily determined by uncertainty in global warming, but rather by regional ocean dynamics.

In stark contrast, we find that weighting regional precipitation trends by how well models reproduce historical global warming does not lead to wide-spread reduction in RMSE (Figure 2c). This finding can largely be attributed to the lack of correlation between *global-mean temperature trends* and *regional-mean precipitation trends* over the 21st century (Figure 2d) outside of the polar regions.

This low degree of correlation suggests that outside the polar regions, uncertainty in regional precipitation is not dominated by the same processes that determine uncertainty in global warming trends. Indeed, the future regional precipitation in polar regions is likely well-correlated with global temperature because it is primarily 'thermodynamically controlled', whereas changes in regional precipitation in the mid-latitudes and tropics are 'dynamically controlled' and as such the non-polar regional precipitation trends cannot be easily linked to global temperature rise (Emori & Brown, 2005). This could also explain why regional RMSE change in future precipitation projections is heterogeneous, while global projections are improved: while models all agree that precipitation increases with temperature (Held & Soden, 2006), they might disagree on the (dynamically controlled) location of precipitation changes. Moreover, owing to dynamical differences of how individual models represent precipitation, regional precipitation patterns may not be robust prior to weighting, meaning the changes after weighting are also not robust.

In any event, the lack of predictive power of historical global temperature anomalies for regional precipitation projections make this metric a poor choice to weight climate models on regional scales (at least, for precipitation projections). As a quantitative example, average RMSE over the contiguous United States (CONUS) in future temperature is reduced by 20%, while average RMSE in future precipitation over the CONUS is increased by 8% (We provide the same calculations for 17 additional regions in the Supporting Information S1 and find similar results.)

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3. Discussion

Our regional analysis shows that care must be taken when applying a historical temperature-based weighting scheme to other climate variables, particularly for regional impact analysis. Weighting regional temperature projections by global historical temperature projections leads to a reduction in bias for most regions. However, the same is absolutely not true when applying these weights to regional precipitation projections. Indeed, we find that many regions actually have a higher RMSE in future precipitation projections using the weighted mean as opposed to the unweighted mean (see blue regions in Figure 2c). It also follows that a global reduction in bias cannot be conflated with a useful, uniform regional reduction in error; indeed, just because error decreases globally does not imply it also decreases in every city or municipality, or even in most cities or municipalities. We conclude that, based on these results, the skill of a temperature-based weighting scheme for global temperature projections cannot be generalized across different climate metrics.

These results have important implications for both climate model practitioners and climate scientists. For practitioners, it is important to take caution when choosing a weighting scheme so as to avoid an overconfident or biased prediction. Naïvely applying a temperature-based weighting method, such as those adopted by the IPCC, to other climate model variables can lead to misleading results and worse regional predictions than using an unweighted mean, as shown in Figure 2. (Note we also explore a historical precipitation-based analog to the IPCC's temperature-based weighting scheme in the Supporting Information S1 and find similar conclusions.) In other words, temperature-based weighting methods cannot, and should not, be considered general for any regional metrics other than temperature projections without verification on a case-by-case basis. Note that applying an unweighted mean in many cases may still not be preferable, and one would ideally use a weighting scheme that is optimized for the region and variable of interest. Our work would suggest that a reliable first-step would be to determine the degree to which a candidate global weighting metric correlates to the regional climate variable of interest (i.e., our calculations in Figures 2b and 2d) to probe if a weighted projection would improve bias in a climate model ensemble. Note that our approach is generalizable to the case where multiple candidate weighting metrics correlate well with a desired climate variable (see the Supporting Information S1).

More work needs to be done to build out robust, bespoke weighting schemes for different metrics of climate change, particularly those that are important for climate impact analysis. Weighting schemes for multi-faceted climate risk assessment, where multiple variables need to be predicted, are likely to be particularly challenging. Different impact variables could correlate well with different weighting metrics, but using different weighting schemes for each impact variable could lead to inconsistent predictions. If it is possible to define a single weighting metric that correlates well with each climate variable of interest, we recommend that weighting schemes are built around this weighting metric. Additional approaches to address this issue have been suggested, such as recent work focusing on choosing climate models based on independence, performance, and spread (Merrifield et al., 2023); expanding on this framework or developing novel approaches would fill a need for model weighting and selection for specific tasks, particularly on regional spatial scales and near-term timescales, where reliable risk assessment is urgently needed (Condon, 2023). Climate scientists should actively engage with climate model practitioners to lend expertise and insight into the best practices for weighting climate ensembles, lest highly consequential decisions be made based on ill-suited (though well-intentioned) weighted climate model ensembles.

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Data Availability Statement

The raw climate model data listed in the Supporting Information S1 is publicly available in a long-term stable repository at https://esgf-data.dkrz.de/search/cmip6-dkrz/. We have compiled a text list of models to download from the above repository. The code and stable repository model download list to reproduce all analysis can be found in McDonnell (2024) and at the following Github link: https://github.com/abigailmcdonnell/model-democracy.

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